

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)
CHARUSAT Campus, Changa – 388 421.

M Sc BIOCHEMISTRY
SEMESTER II EXAMINATION (REGULAR)
BC 708: HUMAN PHYSIOLOGY

Date: 06.05.2011

Time: 10.00 AM to 1.00 PM

Total Marks: 70

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
 2. Attempt all questions.
 3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
 4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
 5. Use of cell phones is strictly prohibited.
 6. Use of non-programmable calculators may be allowed if needed.
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PART A

Total marks: 20

Q1. Choose the correct option and put $\sqrt$ mark in front of it

1. The function of haptoglobin appears to be prevention of loss of
A. Albumin B. Globulin C. Hemoglobin D. None of the above
2. Other name for factor IX is
A. Stuart Prower factor B. Plasma thromboplastin C. Hageman factor D. Christmas factor
3. $\alpha_2\gamma_2$ indicates any one of the following hemoglobin
A. Abnormal Hb B. Fetal Hb C. Adult Hb D. Glycosylated Hb
4. Any one of the following is an extra cellular buffer
A. Phosphate B. Plasma protein C. Bicarbonate D. Hemoglobin
5. Which of the following carries out chemical digestion?
A. Insulin B. Gastrin C. Trypsin D. Secretin
6. The neurotransmitter that directly stimulates the secretions by the gastric glands is
A. Acetylcholine B. Dopamine C. GABA D. None of the above
7. Intrinsic factor is a glycoprotein secreted by the parietal cells that helps in the adequate absorption of vitamin B₁₂.
(True / False)
8. Which of the following is true about the enzyme trypsin
A. It cleaves peptide bond on carboxyl side of aromatic amino acids
B. It cleaves peptide bond adjacent to aromatic amino acids
C. It cleaves peptide bond on carboxyl side of basic amino acids
D. It cleaves peptide bond on carboxyl side of aliphatic amino acids

9. Glucose is reabsorbed in the kidney mainly by
A. Bowman's capsule B. Loop of Henle
C. Proximal Convoluted Tubule D. Distal Convoluted Tubule
10. In the kidneys of mammals, Loop of Henle can be found in
A. Medulla B. Cortex . C. Pelvis . D. Pyramid
11. A severe fall in blood pressure disturbs the function of kidneys and reduces
A. Renal filtration B. Glomerular filtration C. Reabsorption D. Secretion
12. If a man takes in large amount of proteins he is likely to secrete more amount of
A. Urea B. Uric acid C. Glucose D. CO₂
13. Which of the following cells is responsible for myelin formation in the peripheral nervous system?
A. Astrocyte B. Microglial cell C. Schwann cell D. Satellite cell
14. The system that controls smooth muscle, cardiac muscle, and gland activity is the
A. Somatic nervous system B. Autonomic nervous system
C. Sensory nervous system D. None of the above
15. Sarcomeres run from
A. A band to A band B. Z line to Z line C. H zone to H zone D. I band to I band
16. Individual fibers of skeletal muscle have a fine sheath of connective tissue called _____
A. Epimysium B. Perimysium C. Endomysium D. Fascia
17. Which of the following is produced in the adrenal cortex?
A. Male sex hormones B. Aldosterone C. Cortisol D. All of the preceding
18. Any one of the following is a local hormone
A. NO B. Testosterone C. GH D. Insulin
19. Cyclic AMP is best matched with
A. Steroid hormones B. Protein hormones C. muscle cells D. Testosterone
20. An increase in blood glucose and an anti-inflammatory effect are important effects of
A. Epinephrine B. Glucagon C. Cortisol D. Insulin

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Part B

Total Marks: 50

Section I

- Q1.** A Write note on Hemophilia [4]
OR
A Discuss Contact activation pathway [4]
B Discuss on Three RBC indices used to determine the cause of anemia [3]
OR
B Explain the Transport of CO₂ in blood [3]
C A patient is in intensive care because he suffered a severe myocardial infarction 3 days ago. The lab reports the following values from an arterial blood sample: pH = 7.3, HCO₃⁻ = 20 mEq/L, pCO₂ = 34 mm Hg. Diagnose the patient [2]
OR
C Discuss the role of Hemoglobin as buffer [2]
D Give Clinical significance of ceruloplasmin [1]
- Q2.** A Explain the different phases of gastric secretion [4]
OR
A Explain the mechanism of hydrochloric acid secretion in the stomach [4]
B Write a note on the pancreatic digestive enzymes [3]
C Explain the mechanism of sodium absorption through the intestinal epithelium [3]

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Section II

- Q3.** A Differentiate between cortical and juxtamedullary nephron [2]
OR
A Explain Paracellular and transcellular routes of transport [2]
B Write on Structure of glomerular membrane [3]
OR
B Represent transport max for glucose using graph in normal and diabetic conditions [3]
C Define GFR. Describe indices that make up GFR [4]
OR
C Write Significance of macula densa cells [4]
D Enlist the Characteristics of Proximal tubule epithelial cells [1]
- Q4.** A Explain the role of calcium ions in the skeletal muscle contraction [4]
OR
A Explain synaptic transmission [4]
B Discuss the sources of energy for the muscle contraction [3]
C Discuss the functions of non-nervous supporting cells in the nervous system [3]
- Q5.** A Discuss Central Diabetes Insipidus [4]
OR
A Write short note on Grave's disease [4]
B Explain First 2 steps in the synthesis of thyroid hormones [4]
OR
B Explain Activation of insulin receptor [4]
C Give flowchart : Synthesis of progesterone from cholesterol [2]
OR
C Discuss Physiologic effects of eicosanoids [2]